3.14 Tropical Infections

ADULTS

CHAPTER 3.14	TROPICAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Melioidosis *Burkholderia pseudomallei*			
Initial, **Intensive phase** Without complication	Ceftazidime* 2 g IV q6H ± Co-trimoxazole** (weight-based dosing) **>60 kg:** 4 SS tablets PO q12H **OR** 4 ampoules IV q12H **40-60 kg:** 3 SS tablets PO q12H **OR** 3 ampoules IV q12H **<40 kg:** 2 SS tablets PO q12H **OR**		• *Alternate dosing: 2 g loading dose then 8 g daily via continuous infusion. • **Consider adding Co-trimoxazole for severe infection involving the brain, prostate, brain, eyes, testicular, and articular cartilage. • Co-trimoxazole can also be given as oral suspension via nasogastric tube. • **Dosing of Co-trimoxazole (based on weight)** • **>60 kg:** 4 tablets PO q12H **OR** 4 ampoules IV q12H

<table>
<tr><th>CHAPTER 3.14</th><th colspan="3">TROPICAL INFECTIONS</th></tr>
<tr><th rowspan="2">Infections and Likely Organisms</th><th colspan="2">Suggested Treatment</th><th rowspan="2">Notes</th></tr>
<tr><th>Preferred</th><th>Alternative</th></tr>
<tr><td></td><td>2 ampoules IV q12H</td><td></td><td rowspan="2">40 - 60 kg: 3 tablets PO q12H OR 3 ampoules IV q12H
<40 kg: 2 tablets PO q12H OR 2 ampoules IV q12H
Modify treatment according to culture and sensitivity.
Abscesses should be surgically drained.
Duration: 10-14 days.
4 weeks:
Septic joint, deep seated abscess or organ abscesses, extensive lung disease
6 weeks:
Osteomyelitis
8 weeks:
Neurological melioidosis</td></tr>
<tr><td>Initial,
Intensive phase
Present with either of the following conditions:
Neuromelioidosis
Persistent bacteraemia
Requiring ICU stay
Treatment failure</td><td>Meropenem
1 g IV q8H OR
2 g IV q8H (for CNS disease)
±
Co-trimoxazole**
(weight-based dosing)
>60 kg:
4 SS tablets PO q12H OR
4 ampoules IV q12H
40-60 kg:
3 SS tablets PO q12H OR
3 ampoules IV q12H
<40 kg:
2 SS tablets PO q12H OR
2 ampoules IV q12H</td><td></td></tr>
</table>

CHAPTER 3.14	TROPICAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Oral eradication phase	Co-trimoxazole 480 mg tablets **>60 kg:** 4 tablets PO q12H **40 - 60 kg:** 3 tablets PO q12H **<40 kg:** 2 tablets PO q12H	Co-amoxiclav 625 mg tablets **>60 kg:** 3 tablets PO q8H **<60 kg:** 2 tablets PO q8H	• Duration: Minimum 12 weeks • OM and neurological diseases: 6 months. • Use Co-amoxiclav only if resistant or allergic to Co-trimoxazole.
Rickettsia Scrub, Murine, and Epidemic Typhus			
Scrub typhus *Orientia tsutsugamushi* (previously called *Rickettsia tsutsugamushi*)			
Uncomplicated	Doxycycline 100 mg PO q12H for 7-10 days	Azithromycin 500 mg IV/PO q24H for 5-7 days	• **For pregnant patients:** Azithromycin 500 mg IV/PO q24H for 7 days.
Complicated/severe (organ dysfunction)	Azithromycin 500 mg IV q12H on day 1 Then, 500 mg IV q24H for 6 days[1]	**If suspected Doxycycline resistance:** Azithromycin 500 mg IV q12H on day 1	• *In the absence of intravenous Doxycycline for complicated infection.

CHAPTER 3.14	TROPICAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	+ Doxycycline* 200 mg PO q12H on day 1 Then, 100 mg PO q12H for 6 days	Then, 500 mg IV q24H for 6 days **OR** Rifampicin 600 mg PO q24H for 5 days	
Murine typhus			
Rickettsia typhi (transmitted via rat flea)	Doxycycline 100 mg PO q12H for 7 days		• **For pregnant patients:** Chloramphenicol* 12.5 mg/kg (max 4 g daily) IV q6H for at least 7 days. • *Chloramphenicol IV/PO are not currently available in Malaysia.
Epidemic typhus			
Rickettsia prowazekii (transmitted via human louse)	Doxycycline 100 mg PO q12H for 5 days		• De-louse clothing, i.e., discarding, laundering (in hot water), or ironing infested clothing and bedding.
Enteric/Typhoid Fever			
Salmonella typhi	Ceftriaxone 2 g IV q12-	Ciprofloxacin 400 mg IV	• Severe, complicated cases or bacteraemia requires 10-14

CHAPTER 3.14	TROPICAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Salmonella paratyphi A, B, and C	24H ***Once culture is available, to target antibiotics according to sensitivities.*** ***If sensitive,*** Ciprofloxacin 400 mg IV q12H for 7 - 14 days (switch to PO 500 mg q12H as soon as possible) **OR** Ciprofloxacin 500 mg PO q12H for 5-7 days (in mild and uncomplicated disease)	q12H for 7-10 days* **OR** Azithromycin 1 g PO q24H for 5-7 days*	days duration of treatment. • If septic shock or severely ill, use Dexamethasone 3 mg/kg single dose then 1 mg/kg q6H for 8 doses a few minutes before antibiotic. • *If acquired from Indian sub-continent or South East Asia (Indonesia, Thailand, Myanmar) and mild disease, to consider Azithromycin prior to obtaining Fluoroquinolone susceptibility. • *If acquired in the United States, Pakistan or Iraq (where there is high risk of extremely drug resistant *S. Typhi*), may require Azithromycin or Carbapenem or both – refer ID[2].
Non-typhoidal Salmonella			
Bacteraemia	Ceftriaxone 2 g IV q24H till improve clinically then convert to oral ***Once culture is***	**If proven susceptible or allergy to penicillin:** Co-trimoxazole (Trimethoprim	• Look for dissemination (mycotic aneurysm, OM etc). • *Duration:

CHAPTER 3.14	TROPICAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	available, to target antibiotics according to sensitivities. If sensitive, for Ciprofloxacin 400 mg IV q12H for 7 - 14 days* (switch to PO 500 mg q12H as soon as possible).	component) 8-10 mg/kg/day IV/PO in divided doses q8H	• **No extra-intestinal infection, dissemination and immunocompetent:** 14 days • **Extra-intestinal infection (Mycotic aneurysm, osteomyelitis, or immunosuppressed):** 4 – 6 weeks[3] • Risk of dissemination and complicated infections: • Age <12 months or >50 years of age, and immunosuppressed**. • Immunosuppressed** • Patients with HIV infection, especially if poorly controlled • Organ transplant recipients • Patients receiving corticosteroids or other immunosuppressive agents • Patients with cancer or lymphoproliferative disease with current or recent chemotherapy • Patients with sickle cell disease or other hemoglobinopathies • Patients with disorders of the reticuloendothelial

| | | | system, including cirrhosis and chronic granulomatous disease
• Children with severe malarial anaemia and malnutrition
• Patients with atherosclerosis or prosthetic heart valves
• Avoid fluoroquinolones in pregnant women, children. |
| **Gastroenteritis** | Ciprofloxacin 500 mg PO q12H for 7-10 days (14 days if immunocompromised) | | • Usually self-limiting and primary treatment is fluid and electrolyte replacement.
• Antibiotic is warranted for treatment of patients with severe disease and high risk for invasive disease:
• Age > 50 years
• Immunocompromised
• Haemodialysis
• Vascular aneurysm/grafts/ |

CHAPTER 3.14	TROPICAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
			• prosthetic joints • Bacteraemia • Hemoglobinopathies • Hospitalised with fever and severe diarrhoea
Leptospirosis *L. icterohaemorrhagiae*, *L. canicola*			
Mild/Outpatient	Doxycycline 100 mg PO q12H for 5-7 days	Amoxicillin 500 mg PO q8H for 7 days **OR** Azithromycin 1 g IV/PO q24H D1, then 500 mg IV/PO q24H D2 and D3	• Jarisch-Herxheimer reaction can occur post-penicillin.
Severe Leptospiral pulmonary syndrome, multiorgan involvement, sepsis	Benzylpenicillin IV 1.5 MU q6H for 7 days	Ceftriaxone 2 g IV q24H for 7 days	

CHAPTER 3.14	TROPICAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Brucellosis *B abortus* (cow), *B suis* (swine), *B melitensis* (goats), *B canis* (dogs)			
Systemic Disease	Doxycycline 100 mg PO q12H for 6 weeks + Gentamicin 4 - 6 mg/kg IV q24H for 7-10 days[4] (Monitor serum Gentamicin level)	Doxycycline 100 mg PO q12H for 6 weeks + Rifampicin 600 mg PO q24H for 6 weeks	• Relapse rate: 10%, 20-30% (bone involvement), 1% (neurobrucellosis) • Steroids are not recommended. • Ciprofloxacin: active in vitro but clinical response is not good. • Duration: 6 weeks • **For pregnant women:** Rifampicin 600 – 900 mg (15mg/kg) PO q24H for 6 weeks + Co-trimoxazole 480 mg tablets (2 tabs) q12H for 6 weeks (supplemented by Folic or Folinic Acid) • **For pregnant women ≥36 weeks' gestation**: Rifampicin monotherapy until delivery, given risk of neonatal kernicterus with the use of TMP-SMX in the last month of pregnancy[5,6].
Spondylitis, Sacroiliitis	Doxycycline 100 mg PO q12H for 12 weeks	Ciprofloxacin 750 mg PO q12H for 12 weeks	• Duration: 3 months. • **For pregnant women:**

CHAPTER 3.14	TROPICAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	+ Gentamicin 5 mg/kg IV q24H for 1-2 week (Monitor serum Gentamicin level) + Rifampicin 600 - 900 mg PO q24H for 12 weeks	+ Rifampicin 600 - 900 mg PO q24H for 12 weeks	Ceftriaxone 2 g IV q24H for 4 to 6 weeks + Rifampicin 600-900 mg q24H for at least 12 weeks + Co-trimoxazole 2 tablets (480 mg tablet) PO q12H for at least 12 weeks • **Pregnant women ≥36 weeks' gestation:** To omit co-trimoxazole, given risk of neonatal kernicterus. To give ceftriaxone and rifampicin until delivery. Restart o-trimoxazole post-delivery until total 12 weeks[6].
Endocarditis	Doxycycline 100 mg PO q12H for 12 weeks + Rifampicin 600 mg PO q24H for 12 weeks + Gentamicin 5 mg/kg IV q24H for the first 4 weeks[6]	Doxycycline 100 mg PO q12H for 12 weeks + Rifampicin 600 mg PO q24H for 12 weeks + Streptomycin 1 g IM/IV q24H for the first 4 weeks **OR**	• Treatment success defined as an antibody titre <1:60[8]. • Early cardiothoracic referral • Duration of treatment: 6 weeks – 6 months. • Duration of Gentamicin: 2 – 4 weeks. • **Pregnant women:** Ceftriaxone 2 g IV q24H for 4 to 6 weeks + Rifampicin 600 – 900 mg PO q24H at least 12 weeks + Co-trimoxazole 2 tabs (480 mg tablets) PO q12H for at least 12 weeks

CHAPTER 3.14	TROPICAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
		Doxycycline 100 mg PO q12H 3-6 months[8] + Co-trimoxazole 2 tabs (480 mg tablets) PO q12H for 3-6 months[8] + Rifampicin 300 – 600 mg PO q24H for 3-6 months[8]	• **Pregnant women ≥36 weeks' gestation:** Ceftriaxone and Rifampicin until delivery, given risk of neonatal kernicterus with co-trimoxazole use in the last month of pregnancy. Restart Co-trimoxazole post-delivery until total 12 weeks • If unable to tolerate Streptomycin or Gentamicin, to consider Co-trimoxazole. • Monotherapy with Rifampicin, Oxytetracycline, or Doxycycline showed relapse rates of 9-25%; monotherapy with TMP/SMX or Ciprofloxacin has relapse rate of 30% up to 83%, respectively[9].
Neurobrucellosis	Doxycycline 100 mg PO q12H for 12 weeks + Rifampicin 600 - 900 mg PO q24H for 12 weeks + Ceftriaxone 2 g IV q12H for 4-6 weeks	Doxycycline 100 mg PO q12H for 12 weeks + Rifampicin 600 - 900 mg PO q24H for 12 weeks + Co-trimoxazole (Trimethoprim component) 5 mg/kg PO q12H in divided dose for 12 weeks	• Clinical response to Ceftriaxone varies. • Duration: until CSF clears, duration of treatment might need extension up to 4-6 months. • **Pregnant women:** Ceftriaxone 2 g IV q12H for 4 to 6 weeks + Rifampicin 600-900 mg PO q24H for at least 12 weeks + Co-trimoxazole 2 tablets (480 mg tablet) PO q12H at least 12 weeks

CHAPTER 3.14	TROPICAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
			• **Pregnant women ≥36 weeks' gestation:** Ceftriaxone and Rifampicin until delivery, given risk of neonatal kernicterus with co-trimoxazole use in the last month of pregnancy. Restart Co-trimoxazole post-delivery until total 12 weeks[6,7]
Post exposure prophylaxis (PEP)	Doxycycline 100 mg PO q12H + Rifampicin 600 mg PO q24H		• Duration: 3 weeks. • Obtain baseline serum Brucella serology, and at 2, 4, 6, and 24 weeks from exposure. • Weekly active surveillance for fever and symptoms of brucellosis.
Cat scratch disease and other Bartonella infections			
Lymphadenitis	Azithromycin 500 mg PO D1 then 250 mg PO q24H for 4 days	Doxycycline 100 mg PO q12H for 14 days	
Hepatosplenic involvement	Rifampicin 300 mg IV/PO q12H + Azithromycin 500 mg PO		

CHAPTER 3.14	TROPICAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	D1 then 250 mg q24H for 10-14 days [10]		
Neuroretinitis	Doxycycline 100 mg IV/PO q12H + Rifampicin 300 mg IV/PO q12H for 4-6 weeks[10]	Rifampicin 300 mg IV/PO q12H + Azithromycin 500 mg PO D1 then 250 mg q24H for 4-6 weeks **OR** Rifampicin 300 mg IV/PO q12H + Co-trimoxazole 1tab (480 mg tablets) PO q12H for 4-6 weeks[10]	• Adjunctive corticosteroid is recommended. • Prednisolone: 1 mg/kg/day PO for 2 weeks then taper over the following 4 weeks.
Endocarditis	Doxycycline 100 mg PO q12H for 6 weeks	Azithromycin 500 mg PO q24H for 3 months	• For cardiothoracic referral. • Duration:

CHAPTER 3.14	TROPICAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	+ Gentamicin* 1 mg/kg IV q8H for 2 weeks (Monitor serum Gentamicin level) **OR** Rifampicin 300 mg PO q12H for 2 weeks	+ Gentamicin 1 mg/kg IV q8H for 2 weeks (Monitor serum Gentamicin level) **OR** Rifampicin 300 mg PO q12H for 2 weeks	• Regimen with Doxycycline is to be continued for 3 months (if valve not removed) • Regimen with Azithromycin is to be continued for 6 months (if infected valve is surgically resected) • Regime with Azithromycin to be continued for 9 months (if infected valve not resected) • Gentamicin target serum levels: peak 3-4 mg/dL and trough >1mg/dL
Cholera			
Vibrio cholerae	**HYDRATION is primary therapy** Azithromycin** 1 g PO single dose **OR** Doxycycline* 300 mg PO single dose	Erythromycin Ethyl Succinate** 400 mg PO q8H for 3 days **OR** Ciprofloxacin*** 1 g PO single dose	• * Non-tetracycline resistance. • ** Tetracycline resistance and pregnancy: Erythromycin Ethyl Succinate or Azithromycin. • ***Avoid fluoroquinolones in pregnant women[11].
Tetanus			
Clostridium tetani	Metronidazole 500 mg IV q6-8H	Benzylpenicillin 3 MU IV q4H	• Duration: 7 - 10 days.

CHAPTER 3.14	TROPICAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	+ Human Tetanus Immunoglobulin 3000 - 6000 IU/IM single dose + Initiate a full course of tetanus vaccination (3 doses in adults*), avoid giving on the same site as immunoglobulin	**OR** Doxycycline 100 mg PO q12H + Human Tetanus Immunoglobulin 3000 - 6000 IU/IM single dose + Initiate a full course of tetanus vaccination (3 doses in adults*), avoid giving on the same site as immunoglobulin.	• Surgical debridement of infected source tissue. • Penicillin is a GABA antagonist. Thus, it may aggravate the spasms. • **If there is *no previous primary vaccination* series for tetanus, diphtheria, or pertussis:** 1 dose Tdap followed by 1 dose Td or Tdap at least 4 weeks later, and third dose of Td or Tdap 6 to 12 months later (Tdap can be substituted for any Td dose, but preferred as first dose), Td or Tdap every 10 years thereafter.
Malaria *Plasmodium falciparum, Plasmodium vivax, Plasmodium ovale, Plasmodium knowlesi, Plasmodium malariae*			
Complicated Malaria	D1: Artesunate* 2.4 mg/kg IV single dose then 2.4 mg/kg at 12 hours D2: Artesunate* 2.4 mg/kg IV q24H till oral conversion	Artesunate 200 mg/ Mefloquine 440 mg*** PO q24H for 3 days **OR** Quinine*** (10 mg/kg) PO q8H for 7 days	• WHO recommended combination therapy. • **Primaquine (dosed as base):** ◦ Ensure check G6PD status before use. ◦ Contraindicated in pregnant women, children aged < 6

<table>
<tr><th>CHAPTER 3.14</th><th colspan="3">TROPICAL INFECTIONS</th></tr>
<tr><th rowspan="2">Infections and Likely Organisms</th><th colspan="2">Suggested Treatment</th><th rowspan="2">Notes</th></tr>
<tr><th>Preferred</th><th>Alternative</th></tr>
<tr><td></td><td>+
Doxycycline 100 mg PO q12H for 7 days
+
Primaquine 30 mg PO single dose

*Note:
Parenteral Artesunate should be given for a minimum of 24 hours.
Switch to Artemether/ Lumefantrine (Riamet®)** when possible.</td><td rowspan="2">+
Doxycycline 100 mg q12H for 7 days</td><td rowspan="2">months, women breastfeeding child < 6 months.
• **Dosing of Artemether 20 mg / Lumefantrine 120 mg (Riamet®):
• Adult (> 35 kg)
D1: 4 tablets single dose then 4 tablets at 8 hours
D2 to D3: 4 tablets q12H
Total: 24 tablets
• Adult (< 35 kg)
D1: 3 tablets single dose then 3 tablets at 8 hours
D2 to D3: 3 tablets q12H
Total: 18 tablets
• Artemether/Lumefantrine should be taken with fatty meals to improve the oral absorption.
• Pregnancy: Avoid Doxycycline and Primaquine. Relative contraindication of Artemisinin in 1st trimester. Seek expert advice.
• ***Artesunate 200 mg/Mefloquine 440 mg and Quinine tablets are not available at UMMC.</td></tr>
<tr><td>Uncomplicated Malaria
P. falciparum</td><td>Artemether/ Lumefantrine**
+
Primaquine 30 mg PO single dose</td></tr>
</table>

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<tr><td>Uncomplicated Malaria
P. vivax, P. ovale</td><td>Artemether/ Lumefantrine**
+
Primaquine* 15 - 30 mg PO q24H for 14 days</td><td></td><td rowspan="3">• Primaquine (dosed as base):
  • Ensure check G6PD status before use.
  • *For G6PD patients: Primaquine 45 mg PO weekly for 8 weeks.
  • Contraindicated in pregnant women, children aged < 6 months, women breastfeeding child < 6 months.
• **Dosing of Artemether 20 mg / Lumefantrine 120 mg (Riamet®):
  • Adult (> 35 kg)
  D1: 4 tablets single dose then 4 tablets at 8 hours
  D2 to D3: 4 tablets q12H
  Total: 24 tablets
  • Adult (< 35 kg)
  D1: 3 tablets single dose then 3 tablets at 8 hours
  D2 to D3: 3 tablets q12H
  Total: 18 tablets
• Artemether/Lumefantrine should be taken with fatty meal to improve the oral absorption.</td></tr>
<tr><td>Uncomplicated Malaria
P. knowlesi</td><td>Artemether/ Lumefantrine**
+
Primaquine 30 mg PO single dose</td><td></td></tr>
<tr><td>Mixed Malaria Infection</td><td>Treat as Plasmodium falciparum infection</td><td></td></tr>
<tr><td>Malaria Prophylaxis</td><td colspan="3">Refer to Chapter 3.18: Non-surgical Prophylaxis</td></tr>
</table>

CHAPTER 3.14	TROPICAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Soil transmitted helminths			
Ascaris lumbricoides	Albendazole 400 mg PO single dose **OR** Mebendazole* 500 mg PO single dose	Ivermectin* 200 mcg/kg PO single dose	• **Pregnant women:** Pyrantel Pamoate* 11 mg/kg (max 1 g) PO single dose. • WHO allows use of Albendazole in 2nd and 3rd trimester. • *Mebendazole, Ivermectin, and Pyrantel Pamoate are not available in UMMC.
Pinworm *Enterobius vermicularis*	Albendazole 400 mg PO single dose **OR** Mebendazole* 500 mg PO single dose		• Treatment of households warranted in view of high transmission rate. • Repeat dose in two weeks. • *Mebendazole is not available in UMMC.
Whipworm *Trichuris trichiura*	Mebendazole* 500 mg PO q24H for 3 days **OR** Albendazole 400 mg PO q24H for 3 days	Ivermectin* 200 mcg/kg PO single dose	• *Mebendazole and Ivermectin are not available in UMMC.
Hookworm	Albendazole 400 mg PO	Pyrantel Pamoate* 11 mg/kg (max 1 g) PO	• *Mebendazole and Pyrantel Pamoate are not available in

CHAPTER 3.14	TROPICAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Ancylostoma duodenale *Necator americanus*	for 3 days **OR** Mebendazole* 100 mg PO q12H for 3 days	q24H for 3 days	UMMC.
Strongyloides stercoralis	Ivermectin* 200 mcg/kg/day PO for 1-2 days **For hyper infection with sepsis or meningitis in immunocompromised hosts:** Ivermectin* 200 mcg/kg/day PO until stool and/or sputum negative for 2 weeks	**Asymptomatic or intestinal disease:** Albendazole** 400 mg PO q12H for 7 days	• *In immunocompromised, Ivermectin 200 mcg/kg/day PO for 2 days and repeat in 2 weeks. • Severe hyper infection syndrome can occur in patients with depressed cell-mediated immunity, on corticosteroids or HTLV-1 infection. • Larva currens (serpiginous urticarial rash) over the buttock and waist area can occur in chronic low-level auto infection. • *Ivermectin is not available in UMMC. • **Not as effective as Ivermectin.
Intestinal protozoan parasites			
Amoebiasis *Entamoeba histolytica*	Paromomycin* 25 -30 mg/kg/day PO in 3 divided doses for 7 days		• *Paromomycin and Tinidazole are not available in UMMC. • All *E. histolytica* infections need treatment even if there is no

CHAPTER 3.14	TROPICAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	(intraluminal agent) + Metronidazole 400 mg PO q8H for 10 days **OR** Tinidazole* 2 g PO q24H for 3 days		symptom as there is risk of invasive disease and transmission.
Giardiasis *Giardia duodenalis/ lamblia/ intestinalis*	Tinidazole* 2 g PO single dose **OR** Metronidazole 400 mg PO q8H for 5-7 days	Albendazole 400 mg PO q24H for 5 days	• **In immunodeficiency or refractory disease:** Albendazole 400 mg PO q24H + Metronidazole 250 mg q8H for 5 days. • If refractory, work up for hypogammaglobulinemia, IgA deficiency or HIV.
Blastocystis hominis	**If treatment is indicated:** Metronidazole 400 mg PO q8H for 5-10 days		• Difficult to distinguish between commensal or pathogen.
Cryptosporidium *Cryptosporidium parvum*	Nitazoxanide* 500 mg PO q12H for 3		• Responds to treatment of underlying immunosuppression.

CHAPTER 3.14	TROPICAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Cryptosporidium hominis	days **(if immunocompetent)** **OR** 500 – 1000 mg PO q12H for 14 days **(if immunocompromised)**		• In HIV patients: to optimise antiretroviral therapy (ART). • May need Nitazoxanide ± Azithromycin PO for 4-6 weeks in solid organ transplant recipients[12]. • *Nitazoxanide is not available in UMMC.
Cyclospora cayetanensis	Co-trimoxazole 2 tabs (480 mg tablets) PO q12H for 7-10 days	Nitazoxanide* 500 mg PO q12H for 7 days	• **In HIV patient with CD4<200 cells/mm³:** Co-trimoxazole 2 tabs (480 mg tablets) PO q12H for 14 days, followed by secondary prophylaxis using 2 tabs PO q24H 3 times a week.
Cystoisospora belli (previously *Isospora belli*)	**Symptoms in the immunocompetent usually self-limiting but may need antibiotics if symptoms not resolving.** Co-trimoxazole 2 tabs (480 mg tablets) PO q12H for 7-10 days **OR** Ciprofloxacin 500 mg PO		• Opportunistic pathogen in AIDS and other immune deficiencies. • Consider intravenous Co-trimoxazole if severe malabsorption occurs. • HIV/AIDS: May need to increase dose of Co-trimoxazole to 2 - 4 times daily and/or increase the duration to 3 - 4 weeks. • ART is important for treatment success. • **Maintenance therapy in HIV/AIDS:** Co-trimoxazole 2 tabs (480 mg tablets) PO q24H **OR** three

Infections and Likely Organisms	Suggested Treatment		Notes
	Preferred	Alternative	
	q12H for 7-10 days		times in a week until CD4 >200 for 6 months.
Balantioides coli (previously *Balantidium coli*)	Tetracycline* 500 mg PO q6H for 10 days	Metronidazole 600 mg PO q8H for 5 days	• Usually asymptomatic but can progress to fulminating dysentery with multiple intestinal perforations. • *Tetracycline is not available in UMMC.

Rabies

Categories of exposure determines the indicated post-exposure prophylaxis (PEP) procedure for rabies	
Category I	Touching or feeding animals, animal licks on intact skin (no exposure)
Category II	Nibbling of uncovered skin, minor scratches or abrasions without bleeding (exposure)
Category III	Single or multiple transdermal bites or scratches, contamination of mucous membrane or broken skin with saliva from animal licks, exposures due to direct contact with bats (severe exposure)

Post-exposure prophylaxis (PEP) for rabies
1. Thorough **washing and flushing** of all wounds with soap or detergent and copious amounts of immediately, or as soon as possible.
2. Series of **rabies vaccine** administrations promptly started after an exposure (Refer to Table **'Recommendation on PEP and rabies**

immunoglobulin (RIG) for rabies')

Recommended injection sites:

- **Intradermal (ID):**
    - **All age groups:** deltoid, anterolateral thigh or suprascapular regions
- **Intramuscular (IM):**
    - **Adults and children aged ≥2 years:** deltoid area
    - **Children aged <2 years:** anterolateral area of the thigh

Important: Rabies vaccine **should not** be administered IM in the gluteal area.

3. **Rabies immunoglobulin (RIG)** infiltration into and around the wound, promptly after exposure.
    - Administered only once, preferably at initiation of PEP and not > 7 days following the first rabies vaccine dose.
    - RIG is indicated in category III exposures for **immunologically naïve individuals**.
    - Maximum dose for RIG is 40 IU/kg body weight for equine derived RIG (eRIG), and 20 IU/kg body weight for human derived RIG (hRIG).
    - Previously immunised individuals of all ages who have documented evidence of previous PrEP or at least 2 administrations of vaccine for PEP, RIG is not indicated.
    - If an individual has a repeat exposure <3 months after a previous exposure, and has already received a complete PEP, only wound treatment is required; neither vaccine nor RIG is needed.
    - For repeat exposures occurring >3 months after the last PEP, the PEP schedule for previously immunised individuals should be followed; RIG is not indicated
    - Physicians should be prepared to manage anaphylaxis (rare) during RIG administration.
    - Suturing of wounds should be delayed after RIG infiltration, or if unavoidable, sutures should be loose to allow optimal RIG diffusion.

Recommendation on PEP and rabies immunoglobulin (RIG) for rabies			
Categories of Exposure	**Category I**	**Category II**	**Category III**
Immunologically naïve individuals of all age groups	Washing of exposed skin areas No PEP required	Wound washing and immediate vaccination: 2-sites ID on days 0, 3, 7 **OR** 1 site IM on days 0, 3, 7 and between days 14-28 **OR** 2 sites IM on days 0 and 1 site IM on days 7, 21 RIG not indicated	Wound washing and immediate vaccination: 2-sites ID on days 0, 3, 7 **OR** 1 site IM on days 0, 3, 7 and between days 14-28 **OR** 2 sites IM on days 0 and 1 site IM on days 7, 21 RIG is indicated
Previously immunised individuals of all age groups	Washing of exposed skin areas No PEP required	Wound washing and immediate vaccination 1 site ID on days 0, 3 **OR** 4 sites ID on day 0 **OR** 1 site IM on days 0, 3 RIG not indicated	Wound washing and immediate vaccination 1 site ID on days 0, 3 **OR** 4 sites ID on day 0 **OR** 1 site IM on days 0, 3 RIG not indicated

References:

1. George M varghese IV Doxy, azithro or both for severe Scrub Typhus. NEJM 2023: 388:792-803
2. Varghese, George M., et al. "Intravenous doxycycline, azithromycin, or both for severe scrub typhus." New England Journal of Medicine 388.9 (2023): 792-803.
3. Gordon MA, Banda HT, Gondwe M, et al. Non-typhoidal salmonella bacteraemia among HIV-infected Malawian adults: high mortality and frequent recrudescence. AIDS 2002; 16:1633.
4. Solís García del Pozo J, Solera J. Systematic review and meta-analysis of randomized clinical trials in the treatment of human brucellosis. PLoS One. 2012;7(2):e32090. doi: 10.1371/journal.pone.0032090. Epub 2012 Feb 29. PMID: 22393379; PMCID: PMC3290537
5. https://www.moh.gov.my/moh/images/gallery/Garispanduan/Human_Brucellosis.pdf (access 3/7/24)
6. UpToDate: Brucellosis (Date assessed: December 2023)
7. American Academy of Pediatrics. Red Book: 2024-2027 Report of the Committee on Infectious Diseases, 33rd ed, Kimberlin DW, Banerjee R, Barnett ED, et al (Eds), American Academy of Pediatrics 2024
8. Delgado, Victoria, et al. "2023 ESC Guidelines for the management of endocarditis: Developed by the task force on the management of endocarditis of the European Society of Cardiology (ESC) Endorsed by the European Association for Cardio-Thoracic Surgery (EACTS) and the European Association of Nuclear Medicine (EANM)." European heart journal 44.39 (2023): 3948-4042.
9. Lubani MM, Dudin KI, Sharda DC, Ndhar DS, Araj GF, Hafez HA, et al. A multicenter therapeutic study of 1100 children with brucellosis. Pediatr Infect Dis J 1989; 8: 75-8.
10. Sanford Guide App, App Version 6.4.15, last content update 3/5/2024
11. Teh CS, Suhaili Z, Lim KT, Khamaruddin MA, Yahya F, Sajili MH, Yeo CC, Thong KL. Outbreak-associated Vibrio cholerae genotypes with identical pulsotypes, Malaysia, 2009. Emerg Infect Dis. 2012 Jul;18(7):1177-9. doi: 10.3201/eid1807.111656. PMID: 22709679; PMCID: PMC3376796.
12. Legrand F, Grenouillet F, Larosa F, Dalle F, Saas P, Millon L, Deconinck E, Rohrlich PS. Diagnosis and treatment of digestive cryptosporidiosis in allogeneic haematopoietic stem cell transplant recipients: a prospective single centre study. Bone Marrow Transplant. 2011 Jun;46(6):858-62. doi: 10.1038/bmt.2010.200. Epub 2010 Aug 23. PMID: 20729925.